A Thought Experiment on Double-Slit Interference from a Localized Odd-Harmonic Source — Shape Preservation Is Conditional and Fragile, and the Single-Wavelength N=1 Is the Robust Special Case (Alignment Condition, Tolerance Band, Off-axis Scattering)

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*This paper is a **thought experiment (model calculation)**, not a measurement. It contains no claim that overturns established physics (standard wave optics / quantum mechanics). It illustrates and organizes, by exact analysis and numerical computation, the far-field interference when a **localized source** with positional fluctuation is passed through a double slit. We take the preceding position-readout paper [1] as the sole self-reference and limit external citations to established textbook facts (spatial coherence / far-field double slit; the closed form of the odd-harmonic cosine sum).*

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Abstract

The preceding paper [1] showed that when a **single-wavelength** point source has a positional fluctuation $P(y)$, in each trial the far-field double-slit fringe **shifts by the pure geometric path difference while keeping its shape**, and that over repeated trials the distribution of the fringe shift is the push-forward of P, which **preserves the shape** because the map is linear in the paraxial regime ($\cos^2$ if $P = \cos^2$).

This paper extends that to a **localized source wave**. We represent the localization by a constant-amplitude odd-harmonic sum (the isolated peak wave S_N) and pass it through the double slit. The results are threefold.

1. **Shape preservation (when aligned).** Although S_N is a sum of odd harmonics, it appears in the far field as a single, sharply localized fringe **without changing shape**. This holds only when the source–slit diagonal distance $r_k = \sqrt{L^2 + (W/2)^2}$ is "aligned" to an integer or half-integer multiple of the fundamental wavelength λ_0:
$$\sqrt{L^2 + (W/2)^2} = \frac{m}{2}\,\lambda_0 \qquad (m \in \mathbb{Z}).$$
2. **Sharp alignment condition.** Alignment is not a knife edge but a band of width $\sim 1/(2N)$, narrowing as the highest odd-harmonic order N grows. A **slight difference** in λ_0 (about 3% in our example) decides whether interference forms or scatters. The central-alignment tolerance band itself is **independent of** W/L; the effect of W appears as off-axis fragility (§3.3, §4.3).
3. **Inheritance and fragility of the $\cos^2$ readout.** An aligned localized source still traces $\cos^2$ in the fringe-shift distribution under positional fluctuation (inheriting the push-forward of [1]). But off-axis ($y \neq 0$) the source–slit path difference becomes unequal on the two arms, the localization alignment breaks slightly, and the fringe peak shifts **downward** from $\cos^2$. This departure grows with $|y|$ and N. The single wavelength ($N = 1$) has none of this fragility and is **unconditional**; running our method at $N = 1$ under the same conditions reproduces [1] to machine precision (verified in §5).

The net new content of this paper is a negative result: localization ($N \geq 2$) does not improve the simple push-forward of [1]; it adds an **alignment constraint** (3.4) and **off-axis fragility** that $N = 1$ did not have. Shape preservation is real but **conditional and fragile**, and $N = 1$ is the robust special case. For the broader program from localized waves toward Born statistics, this fragility itself is an important finding. These are not the derivation of a new probability law but illustrations of the change of variables that maps the input distribution onto the observed statistic (fringe shift), and of the alignment condition / sensitivity / fragility of the interference of the odd-harmonic localized wave. This paper is a thought experiment; it organizes established physics rather than overturning it. We note that the visibility loss from **intensity accumulation** of an extended source (spatial coherence / van Cittert–Zernike theorem [2,3]) is a different observation mode from the one of this paper (the push-forward of the fringe shift) (§6.1).

1. Introduction

Double-slit interference is a basic stage in wave optics and quantum mechanics. The preceding paper [1] asked, for a **stationary point source with positional fluctuation** (single wavelength λ_0), what statistics appear in the fringe shift over repeated observation, and showed: each trial shifts the same-shape fringe by the geometric path difference; the shift is a nearly linear map of the source position; hence drawing the source position repeatedly from $P(y)$, the fringe-shift distribution is the push-forward of P, preserving its shape in the paraxial regime.

Here we replace the source of that setting by a **localized wave**. That is, instead of the single-wavelength cos of a point source, we take as the source waveform a wave S_N that is sharply localized at the center by superposing constant-amplitude odd harmonics. We ask two things: (i) does the localized wave, which is a sum of odd harmonics, localize in the far field **while preserving its shape**, and under what conditions; (ii) when that localized source has positional fluctuation, is the $\cos^2$ readout of [1] **inherited, or does it break down**.

We take the same geometry as [1] ($L = 10$, $W = 5$, λ_0 the fundamental wavelength). Everything is an exact computation on the specified configuration; this is a thought experiment. The notation and the closed form of the cosine sum follow a standard analysis text [4]; the far-field double slit and spatial coherence follow standard optics texts [2,3].

2. The localized odd-harmonic source

2.1 The isolated peak wave S_N

On the half-wave phase interval $\varphi \in [-\pi/2, \pi/2]$, define the constant-amplitude odd-harmonic cosine sum

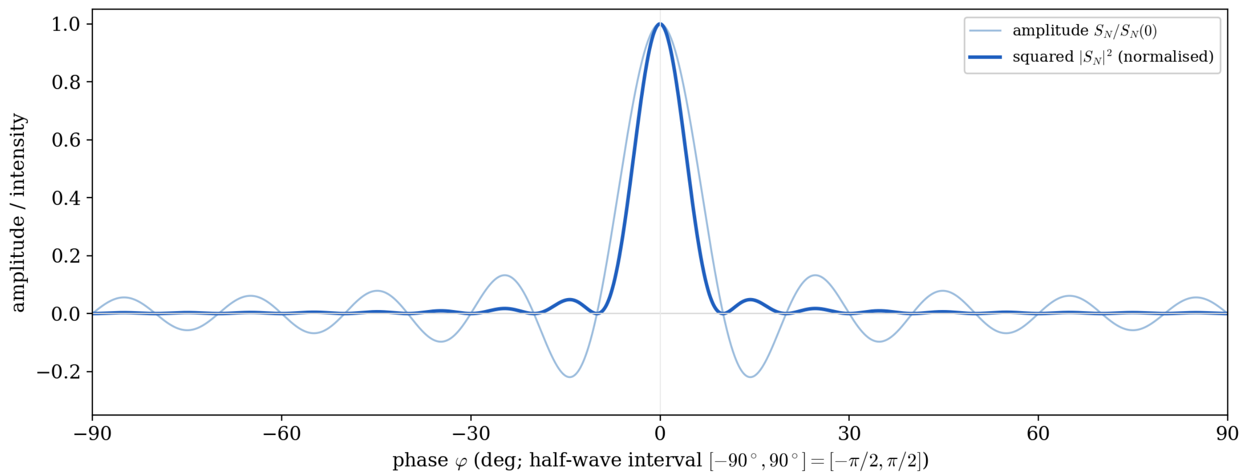
$$S_N(\varphi) = \sum_{m=0}^{(N-1)/2} \cos\big((2m+1)\varphi\big) = \frac{\sin\big((N+1)\varphi\big)}{2\sin\varphi} \tag{2.1}$$

(N the highest odd-harmonic order, odd; the right-hand side is the closed form of the cosine sum of an arithmetic progression [4]). S_N has a central peak $S_N(0) = (N+1)/2$ at $\varphi = 0$ and is zero at the ends $\varphi = \pm\pi/2$ — an "isolated peak wave." For the observable we use the non-negative squared amplitude $|S_N|^2$, whose central peak width narrows as $1/(N+1)$ (Fig 1).

Fig 1: The localized source wave $S_N(\varphi) = \sum_m \cos((2m+1)\varphi)$ ($N = 17$, 9 odd harmonics). Thin line: normalized amplitude $S_N/S_N(0)$; thick line: normalized squared amplitude $|S_N|^2$. A single sharp localized peak at the center, zero at the ends $\pm 90^\circ$. The fundamental frequency $\nu = 1$ and wavelength λ_0 are unchanged by the harmonic carving ($\gcd(1, 3, \ldots, N) = 1$).

2.2 The fundamental wavelength λ_0 (cage) and the harmonics

We call the fundamental wavelength λ_0 (the longest wavelength = the $n = 1$ wavelength) the "cage." The harmonic wavelengths are $\lambda_n = \lambda_0/n$ ($n = 1, 3, \ldots, N$), and λ_0 gives the spatial period of the localized wave. $\nu = 1, \lambda_0$ is a relative normalization; the odd harmonics only change the sharpness of the localization (relative width $1/(N+1)$) and do not change the fundamental frequency / wavelength.



Localized odd-harmonic source wave $S_N(\varphi) = \sum_m \cos((2m+1)\varphi)$, $N = 17$ (9 harmonics). Central peak $S_N(0) = (N+1)/2$, zeros at $\pm 90°$; the squared amplitude is a single localized peak (width $\sim 1/(N+1)$).

Figure 1: Fig 1 localized odd-harmonic source wave S_N (N=17)

3. Alignment condition and shape preservation

3.1 Far-field double slit (exact)

Let the source be $S = (-L, y)$ and the slits $A_1 = (0, +W/2)$, $A_2 = (0, -W/2)$. The screen point is parametrized by the far-field diffraction angle θ_s ($s = \sin\theta_s$). The wavelength is each λ_n everywhere (no motion assumption, no Doppler). The phase of each harmonic n and arm k is the sum of the source-side geometric path and the far-field screen-side path difference:

$$\Phi_k^{(n)}(s; y) = \frac{2\pi n}{\lambda_0}\Big[r_k(y) - y_{\text{slit},k}\, s\Big], \qquad r_k(y) = \sqrt{L^2 + (y - y_{\text{slit},k})^2} \tag{3.1}$$

(the harmonic version of the formula in [1]; $2\pi/\lambda_n = 2\pi n/\lambda_0$). The total complex amplitude and intensity (exact Born form via complex conjugation) are

$$\psi(s; y) = \sum_n \Big[e^{i\Phi_1^{(n)}} + e^{i\Phi_2^{(n)}}\Big], \qquad I(s; y) = \big|\psi(s; y)\big|^2. \tag{3.2}$$

3.2 The alignment condition for the centered source $y = 0$

For the centered source, $r_1 = r_2 = r_k \equiv \sqrt{L^2 + (W/2)^2}$, and (3.2) becomes

$$\psi(s; 0) = \sum_n e^{\,i(2\pi n/\lambda_0)\, r_k} \cdot 2\cos\Big(\frac{2\pi n}{\lambda_0}\frac{W}{2}\, s\Big). \tag{3.3}$$

The source-side phase $e^{i(2\pi n/\lambda_0) r_k}$ is common to both slits but differs per harmonic n. The condition that it align over all odd n (factor out as a common factor) is that $2\pi n\, r_k/\lambda_0$ be the same value (mod 2π) independent of n, i.e.

$$\boxed{\frac{r_k}{\lambda_0} = \frac{m}{2} \quad (m \in \mathbb{Z}), \quad \text{i.e.} \quad \sqrt{L^2 + (W/2)^2} = \frac{m}{2}\,\lambda_0} \tag{3.4}$$

(m even gives phase $+1$, m odd gives $(-1)^n = -1$; both align identically after squaring). Physically it means "the source–slit diagonal distance is an integer/half-integer number of fundamental wavelengths, i.e. the slit sits on a spike (node) of the localized wave."

When aligned, (3.3) becomes $\psi(s;0) = (\text{common factor}) \cdot 2\, S_N(\theta)$, $I = 4\, S_N(\theta)^2$ ($\theta = \pi W s/\lambda_0$), so **the screen interference intensity is the square of the localized wave itself**. Since $|S_N|^2$ has period 180° in θ, this is strictly **the central one of a periodic train of fringes** (the physical screen $|s| \le 1$ can carry several per side), whose center appears, although a sum of odd harmonics, as a single sharply localized fringe **without shape change** (Fig 2). Note that $4S_N^2$ is the **instantaneous coherent image** of $\omega, 3\omega, 5\omega, \ldots$ phase-locked (a frequency comb), including inter-harmonic cross terms. Under time-averaging by a slow detector the cross terms drop and it reverts to a sum of single-harmonic fringes; this paper treats the instantaneous image as a phase-domain thought experiment.

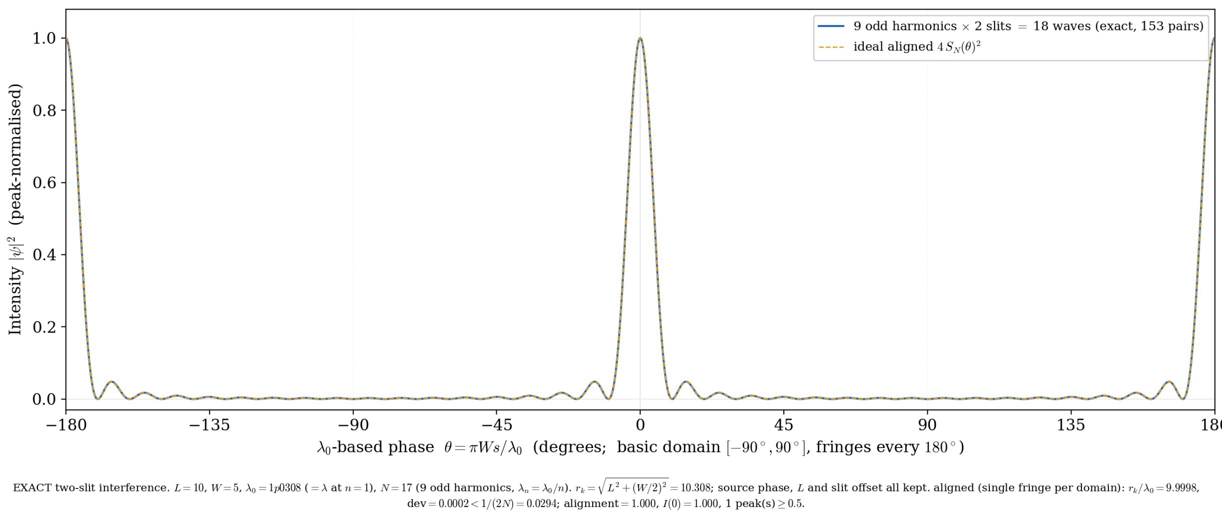


Figure 2: Fig 2 aligned shape-preserving interference (L=10, W=5, 0=1.0308, N=17)

Fig 2: Far-field intensity of the aligned centered source ($y = 0$) ($L = 10$, $W = 5$, $\lambda_0 = 1.0308$, $N = 17$, $r_k/\lambda_0 = 10.000$). 18 waves (9 odd harmonics × 2 slits) are summed exactly and squared. A single sharply localized fringe at the center (the central one of a period-180° train; several lie within $|s| \le 1$), $I(0) = 1$. Dashed: theory $4S_N(\theta)^2$, machine-precision agreement. Horizontal axis: λ_0-based phase $\theta = \pi W s/\lambda_0$.

3.3 Sensitivity: the tolerance band $\sim 1/(2N)$ and where the W-dependence lives

Alignment is a band, not a knife edge. The phase error grows as $2\pi N \cdot \delta(r_k/\lambda_0)$ at the highest harmonic, so the tolerance is

$$\left| r_k/\lambda_0 - \tfrac{m}{2} \right| \;\lesssim\; \frac{1}{2N}, \tag{3.5}$$

narrower as N grows. For our geometry ($L = 10$, $W = 5$), $r_k = \sqrt{106.25} = 10.3078$. With $\lambda_0 = 1$, $r_k/\lambda_0 = 10.308$, whose distance from the nearest half-integer 10.5 is 0.192. This fits the band (3.5) only for $N < 1/(2 \cdot 0.192) \approx 2.6$, i.e. **only $N = 1$ among odd N**. By contrast $\lambda_0 = 1.0308$ ($r_k/\lambda_0 = 10.000$, a mere $\approx 3\%$ change of λ_0) aligns comfortably even at $N = 17$ (tolerance ± 0.029). **A slight difference in λ_0 decides interference vs scattering.**

The required wavelength-matching relative precision is of order $\sim 1/(2N \cdot r_k/\lambda_0)$ ($r_k/\lambda_0 \approx 10$, so about 0.3% at $N = 17$, 0.05% at $N = 99$). Note that scaling the whole geometry while keeping the ratio W/L fixed leaves r_k/λ_0 unchanged, so a bad ratio cannot be fixed by scaling ($r_k/L = \sqrt{1 + (W/2L)^2}$ is generally irrational, but this holds regardless of how large W/L is). To satisfy alignment, tune the two free parameters L and

λ_0 to (3.4). **The central-alignment tolerance band (3.5) is itself independent of W/L, equal to** $\sim 1/(2N)$; the effect of W appears not in central alignment but as **off-axis fragility (when the source leaves the center)** (quantified in §4.3).

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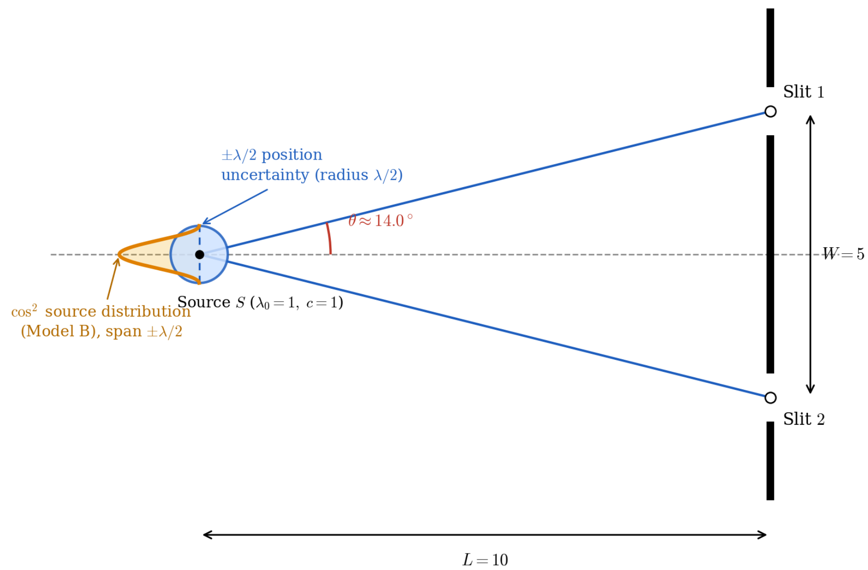
4. The $\cos^2$ readout under positional fluctuation

4.1 The fluctuation setting (identical to [1])

We assign the source position y a fluctuation and take, as in [1], the transverse-position probability distribution

$$P(y) = \cos^2\Big(\frac{\pi y}{\Delta\lambda}\Big), \qquad y \in \Big[-\frac{\Delta\lambda}{2}, \frac{\Delta\lambda}{2}\Big] \tag{4.1}$$

(central peak, zero at the ends; full width $\Delta\lambda$, half-range $\Delta\lambda/2$; $\Delta\lambda = \lambda_0$ corresponds to the $\pm\lambda/2$ of [1]). The source position is quasi-static within a trial and re-drawn between trials according to $P(y)$. The experiments so far (§3, no fluctuation) correspond to $\Delta\lambda = 0$.



Enlarged source-to-slit region ($L = 10$, $W = 5$; screen removed). The source position carries a $\pm\lambda/2$ uncertainty (blue circle, radius $\lambda/2$). The $\cos^2$ source-position distribution (Model B; orange) is drawn along the diameter through the centre, parallel to the screen, spanning $\pm\lambda/2$ (peak at the centre, zero at $\pm\lambda/2$).

Figure 3: Fig 3 source with positional fluctuation (the setup of [1])

Fig 3: Magnified view of the source–slit region (from [1], $L = 10$, $W = 5$). The source position has an uncertainty of radius $\sim \lambda/2$, and along the axis parallel to the screen rides the distribution $P(y) = \cos^2$ (central peak, zero at the ends). This paper replaces this source by the localized wave S_N.

4.2 Push-forward of the fringe shift (exact computation)

For each source position y we compute (3.2) exactly (off-axis $r_1(y) \neq r_2(y)$, keeping all source-side phases), weight by $P(y)$, and normalize by the central peak at $y = 0$. The central peak of the localized fringe stands where the difference phases of all harmonics align, $\Delta r(y) - Ws = 0$, at the position

$$\Delta r(y) = r_1(y) - r_2(y) \;\approx\; -\frac{W}{L}\, y \text{ (length, linear in the paraxial regime)}, \qquad u(y) = \frac{2\pi}{\lambda_0}\,\Delta r(y) \;\approx\; -\frac{2\pi W}{L\lambda_0}\, y \text{ (phase)} \tag{4.2}$$

which is identical to the single-wavelength fringe shift of [1]. Hence the fringe-shift distribution is the push-forward $\rho(u) = P(y(u))|dy/du|$ of P, preserving the $\cos^2$ shape in the paraxial regime.

4.3 Results: inheritance, fragility, and scattering

(a) $N = 1$ **(single wavelength, reproducing [1]).** With only one harmonic, the source-side phase $e^{i\kappa_1 r_k}$ is a mere overall phase and cancels in $|\psi|^2$. Hence the alignment condition (3.4) is **not needed**: whatever L, W, λ_0, the fringe shifts in shape and the peaks trace $\cos^2$. Running our method at $N = 1$ under the same conditions ($L = 10$, $W = 5$, $\lambda_0 = 1$, $\Delta\lambda = 1$) agrees with [1]'s figure (fig_decomposition_static) in **all numbers and the whole curve to machine precision** ($\leq 2 \times 10^{-16}$) (§5).

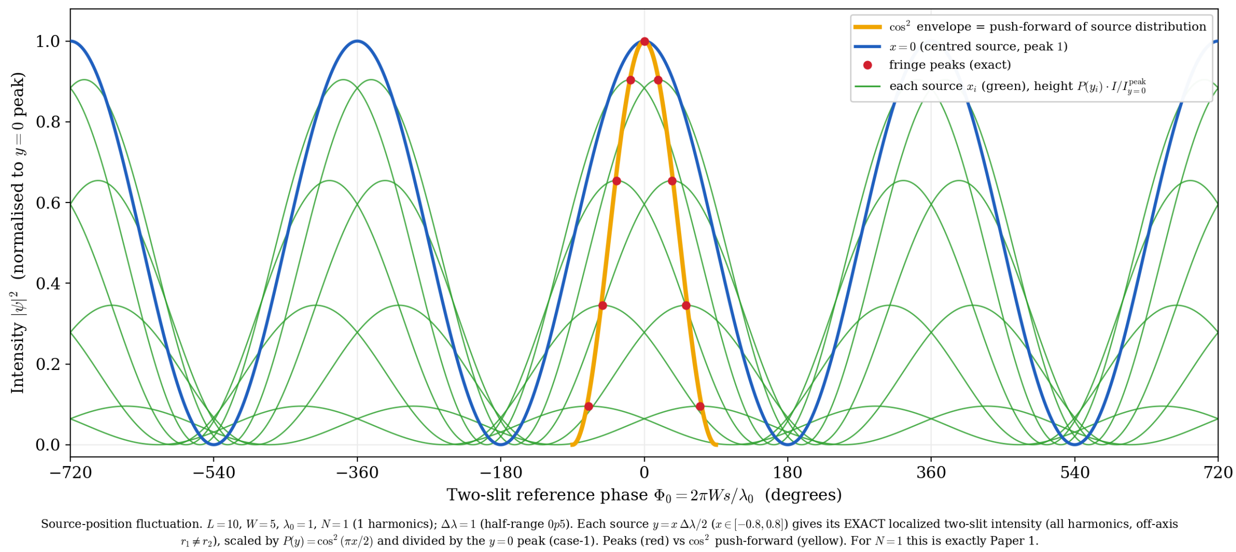


Figure 4: Fig 4 N=1 positional-fluctuation readout (machine-precision match with [1])

Fig 4: $N = 1$, $L = 10$, $W = 5$, $\lambda_0 = 1$, $\Delta\lambda = 1$. *The exact two-slit intensity of each source* $y = x\,\Delta\lambda/2$ ($x \in [-0.8, 0.8]$) *is weighted by* $P(y) = \cos^2(\pi x/2)$ *and normalized by the* $y = 0$ *peak. The fringe peaks (red) trace the* $\cos^2$ *push-forward (yellow). This is exactly [1], and serves as the verification that the base of this paper agrees with [1].*

(b) $N = 17$**, misaligned** ($\lambda_0 = 1$)**.** As in §3.3, with $\lambda_0 = 1$, $N = 17$ is outside the tolerance band, and **the localization fails already at** $y = 0$ (central coherence $\approx 25\%$). The fringe splits into multiple peaks and the peaks depart greatly from $\cos^2$ (Fig 5). This is not a fluctuation effect; it shows that **a misaligned localized source does not interfere/localize at all.**

Fig 5: $N = 17$, $\lambda_0 = 1$ *(misaligned,* $r_k/\lambda_0 = 10.308$*). Even at* $y = 0$ *the central peak is only about* 25% *of the ideal* $(N+1)^2$*, and the fringe peaks of each source depart greatly from* $\cos^2$ *(yellow) — scattering. Same geometry and fluctuation as Fig 4, but a mere* $\approx 3\%$ *difference in* λ_0 *breaks the interference.*

(c) $N = 17$**, aligned** ($\lambda_0 = 1.0308$)**.** With λ_0 at the aligned value, $y = 0$ is fully coherent (center $= (N+1)^2 = 324$). The sharp localized fringe shifts in shape, and the peaks **nearly** trace $\cos^2$ (Fig 6). But off-axis $r_1 \neq r_2$ breaks the alignment slightly and the peak shifts **downward** from $\cos^2$. This departure splits into two heterogeneous contributions:

- **(i) The benign push-forward nonlinearity** (common to all N, geometric): from the non-paraxial nature of the map $u(y)$; the departures of $N = 1$ ($-0.0055 \sim -0.027$) are this (already in [1]).

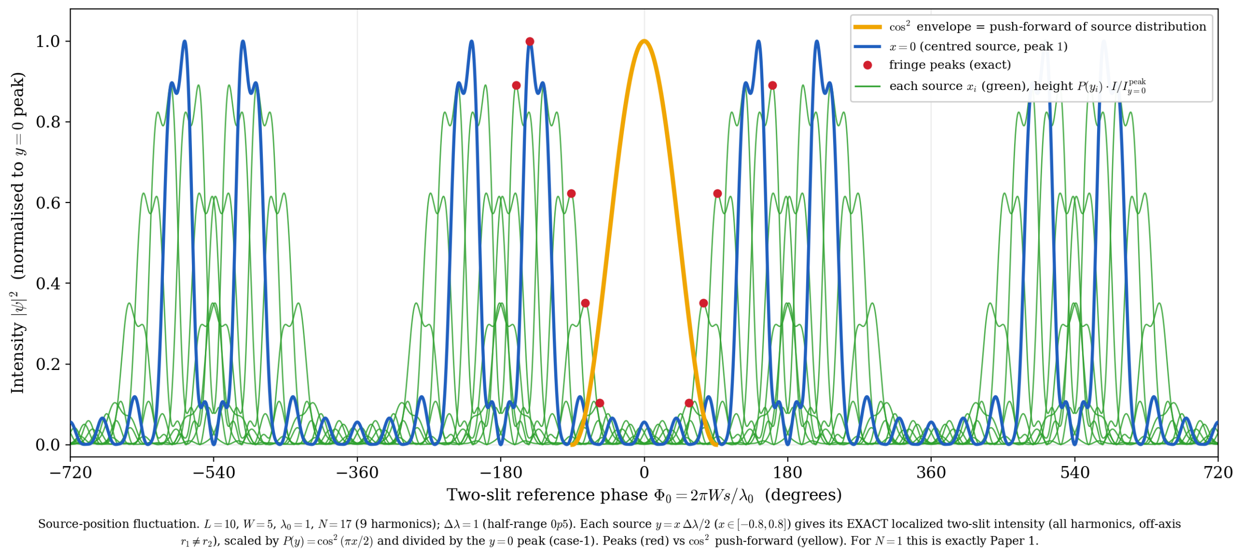


Figure 5: Fig 5 N=17, 0=1 (misaligned): scattering

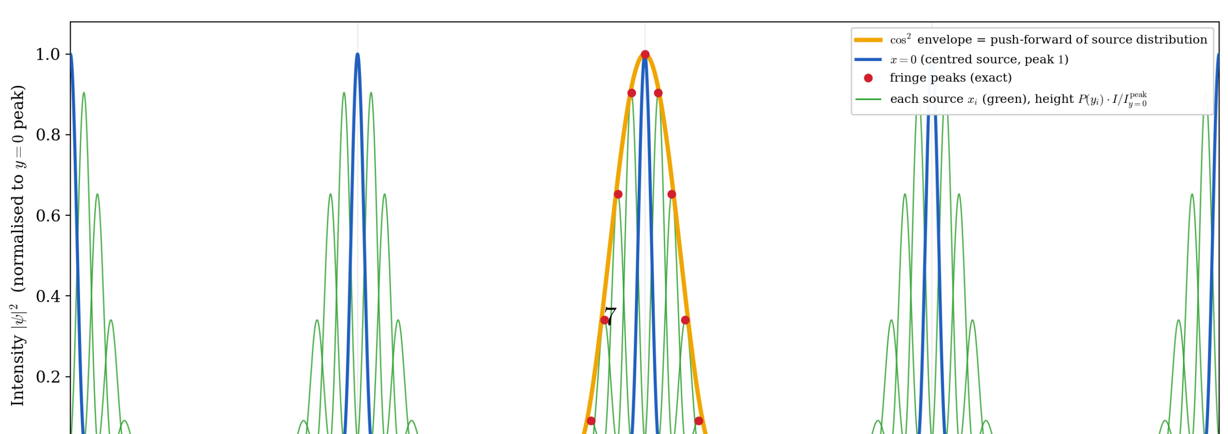
- **(ii) The excess from off-axis scattering** (newly appearing for $N \geq 2$): off-axis $r_1 \neq r_2$ breaks the localization alignment and the localized peak itself drops.

In the table below, the excess by which the $N = 17$ departure exceeds $N = 1$ is this new (ii) scattering (strictly, $N = 1$ is $\lambda_0 = 1$ and $N = 17$ is $\lambda_0 = 1.0308$, slightly different geometries so the (i) component is not exactly equal, but the comparison is broadly valid). For instance at $x = 0.8$ ($y = 0.4$) the peak is about 95% of $P(y)$ (about 5% scattering loss), and the excess grows with $|y|$.

This off-axis scattering depends on W. The rate of path imbalance at the center is $\left|dr_1/dy\right|_{y=0} = (W/2)/r_k \propto W$, so the larger W is, the faster the path leaves the spike (node) when the source leaves the center, and the more scattering. **This is the "true W-dependence" foretold in §3.3**, appearing in the off-axis fragility, not in the central-alignment tolerance band (W-independent).

x	peak height	$\cos^2$ envelope value†	departure ($N = 17$ aligned)	ref: departure ($N = 1$)
0.2	0.9045	0.9153	−0.011	−0.0055
0.4	0.6529	0.6899	−0.037	−0.0178
0.6	0.3404	0.4001	−0.060	−0.0271
0.8	0.0909	0.1442	−0.053	−0.0233

† The "$\cos^2$ envelope value" is the **envelope at the peak position** p_x, $\cos^2(\pi p_x/180^\circ)$, not the weight $\cos^2(\pi x/2)$ (e.g. at $x = 0.2$ the envelope value is 0.9153, while the weight $\cos^2(\pi x/2) = 0.9045$). The "departure" in the table is **peak height − envelope value** (envelope basis). The "about 95% of $P(y)$ (scattering loss 5%)" in the text is on the **weight** $P(y)$ **basis** and is the indicator of the pure scattering (ii). The two have different bases.



N	$\text{peak}_{y=0}$	ideal $(N+1)^2$	alignment	verdict
1	4.0	4	100%	clean
3	8.36	16	52%	scattered
5	14.97	36	42%	scattered
9	27.96	100	28%	scattered
17	80.43	324	25%	scattered

At $\lambda_0 = 1$, only $N = 1$ is aligned; all $N \geq 3$ scatter. Alignment is decided not by N itself but by whether r_k/λ_0 falls in the band (3.5).

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5. Verification ($N = 1$ agrees with [1] exactly)

We directly confirmed that the computational base of this paper agrees with [1]. Running our method at $N = 1$ with $L = 10$, $W = 5$, $\lambda_0 = 1$, $\Delta\lambda = 1$ (running the standard program itself at the same parameters) and comparing with the output of [1]'s standard program (fig_decomposition_static):

- For all 9 source points the fringe peak position, peak height, $\cos^2$ value, and departure **agree in every row**.
- Over the whole 16001-point curve, the **maximum absolute difference is** 2.2×10^{-16} (machine precision).
- The peak value at the center $y = 0$ is 4.0 ($= 2^2$, the two-slit maximum), confirming that the "$y = 0$-peak normalization" of this paper is exactly equivalent to the per-fringe normalization of [1] at $N = 1$ (because the two-slit peak is independent of the source position y).

Thus $N = 1$ is a special limit of this paper, and [1] is that case.

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6. Discussion

6.1 Distinction from spatial coherence (van Cittert–Zernike)

Accumulating the fringes from an extended / multi-point source **on an intensity basis** gives a visibility equal to the Fourier transform of the source brightness distribution: the larger the source, the fainter the fringes (spatial coherence, van Cittert–Zernike theorem [2,3]). This corresponds to observation mode (b) of [1] (intensity accumulation $\to$ visibility loss, not the shape of the input distribution), an established result.

What this paper (and [1]) treats is a different mode (a): reading the fringe **shift** each trial and histogramming over repetitions gives the push-forward of the source **position** distribution, which **preserves the shape**. Visibility loss (mode b / vCZ) and shape preservation (mode a / push-forward) are **different observables** and must not be conflated. The novelty here is to extend this mode (a) to a **localized odd-harmonic source** and to show that shape preservation comes with an **alignment condition (3.4) and a tolerance band** $\sim 1/(2N)$, i.e. a sensitivity to λ_0 — an aspect absent from vCZ.

6.2 Scope (and non-claims)

What this paper shows is limited to: (i) the alignment condition (3.4) and its sensitivity (3.5) under which a localized odd-harmonic source localizes in the double slit while preserving shape; (ii) that an aligned localized source inherits the $\cos^2$ readout under positional fluctuation but the edges shift downward by off-axis scattering; (iii) that $N = 1$ has none of this fragility and agrees with [1] exactly. These are **illustration and organization** by exact computation, not the derivation of a new probability law. The $\cos^2$ (Born form) that appears is the push-forward of the input $P(y) = \cos^2$; we do not ask the origin of the probability interpretation, the squaring rule, or randomness. We do not claim "derivation of the Born rule" or "solution of the measurement problem."

6.3 Implications of the sensitivity (limited remark)

That a slight difference in λ_0 ($\sim 1/(2N)$) decides interference vs scattering, that the required precision tightens as N (the sharpness of localization) grows, and that the larger W/L the greater the off-axis (under-fluctuation) fragility (**while the central-alignment tolerance band itself is independent of W/L**), are consequences of exact computation for our configuration. Between resolution (sharpness of localization $\propto N$) and robustness (alignment / fluctuation tolerance) there is a trade-off controlled by N. We give no physical interpretation beyond this observation.

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7. Conclusion

For far-field double-slit interference from a **localized odd-harmonic source** with positional fluctuation, exact computation on the same geometry as [1] shows the following.

1. Representing the localization by the odd-harmonic sum S_N, the far field gives a single, sharply localized fringe at the center while preserving shape. But this requires the alignment condition $\sqrt{L^2 + (W/2)^2} = (m/2)\lambda_0$ that the diagonal distance be an integer/half-integer multiple of the fundamental wavelength.
2. Alignment is a band of width $\sim 1/(2N)$, narrower as N (the sharpness of localization) grows. A $\approx 3\%$ difference in λ_0 separates interference from scattering. The central-alignment tolerance band itself is independent of W/L, but the larger W/L the greater the off-axis (under-fluctuation) fragility.
3. An aligned localized source still traces $\cos^2$ in the fringe-shift distribution under positional fluctuation (inheriting the push-forward of [1]), but off-axis scattering shifts the edges downward from $\cos^2$. The single wavelength $N = 1$ has none of this fragility and is unconditional; running this paper at $N = 1$ under the same conditions agrees with [1] to machine precision.

In short, the net finding of this paper is that **localization ($N \geq 2$) does not improve the push-forward of [1]; it adds an alignment constraint and off-axis fragility** — i.e. **shape preservation is conditional and fragile, and $N = 1$ is the robust special case.** This is a negative result — that localization complicates and weakens the readout — rather than a positive "shape preservation," and for the program from localized waves toward Born statistics this fragility is the important implication.

This is not the derivation of a new probability law but an extension of the input-distribution push-forward ([1]) to a localized source, illustrating and organizing the alignment condition / sensitivity / fluctuation tolerance (and its fragility) of the interference. It is a different observation mode from the visibility loss of an extended source (spatial coherence / vCZ [2,3]). This paper is a thought experiment; it organizes established physics rather than overturning it.

References

[1] Noriaki Kihara, "A Thought Experiment on Double-Slit Interference from a Source with Positional Fluctuation — Push-forward of the Source-Position Distribution to the Fringe-Shift Distribution (Shape Preservation)," Zenodo, v0.3 (2026), Concept DOI: 10.5281/zenodo.21035808 / Version DOI: 10.5281/zenodo.21035809 (the starting point and sole self-reference).

[2] M. Born and E. Wolf, *Principles of Optics*, 7th (expanded) ed. (Cambridge University Press, 1999) (far-field double slit, partial coherence, van Cittert–Zernike theorem).

[3] L. Mandel and E. Wolf, *Optical Coherence and Quantum Optics* (Cambridge University Press, 1995) (spatial coherence and the van Cittert–Zernike theorem).

[4] T. Takagi, *Kaiseki Gairon* (Introduction to Analysis), 3rd rev. ed. (Iwanami Shoten, 1961) (trigonometric series, the closed form of the cosine sum of an arithmetic progression, the Dirichlet kernel).

Appendix: Reproduction code

All results are reproduced by the following exact-computation scripts (all angles and paths derived from L, W, λ_0; no constants placed arbitrarily; no motion assumption or Doppler).

- Interference / fluctuation engine: `fig_oddharm_interference.py`
  - Aligned shape-preserving interference (Fig 2): `--L 10 --W 5 --lam0 1.0308 --N 17 --halfdeg 180`
  - Positional-fluctuation readout (Figs 4–6): add `--dlam 1` with `--N 1` ($\lambda_0 = 1$) / `--N 17` ($\lambda_0 = 1$) / `--N 17 --lam0 1.0308`
- Localized source wave S_N (Fig 1): `make_paper2_localized_wave.py`
- Standard program of [1] (basis of the verification; the setup of Fig 3): `fig_decomposition_static.py`, `fig_setup_source_uncertainty.py`

The fluctuation mode computes (3.2) exactly at each source $y = x\,\Delta\lambda/2$ ($x \in [-0.8, 0.8]$) over all harmonics, off-axis ($r_1 \neq r_2$), with the source-side phase, weights by $P(y) = \cos^2(\pi x/2)$, and normalizes by the central peak at $y = 0$ (case-1 normalization). At $N = 1$ this agrees with [1] to machine precision (§5).